The EM algorithm

Latent variables and maximum likelihood

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Latent variables

A simple generative model with latent variable:

$$Z \sim p_\theta(z), \qquad X \mid Z \sim p_\theta(x \mid z).$$

$$p_\theta(x, z) = p_\theta(x \mid z) p_\theta(z).$$

This is a generative description: first generate Z, then generate X conditionally on Z.

We observe x, but not z.

$$p_\theta(x) = \int p_\theta(x, z)\, dz.$$

Maximum likelihood

For observed data $x = (x_1, \ldots, x_n)$,

$$\ell(\theta) = \sum_{i=1}^{n} \log p_\theta(x_i), \qquad \widehat{\theta} \in \arg\max_\theta \ell(\theta).$$

- Treat the observations as fixed.
- Optimize over the unknown parameter θ.

$\widehat{\theta}$ is the Maximum Likelihood Estimator (MLE).

Two likelihoods

Observed data

$$\ell(\theta) = \log \int p_\theta(x, z)\, dz.$$

Complete data: if z were known

$$\ell_c(\theta; x, z) = \log p_\theta(x, z).$$

EM uses the complete-data structure to optimize the observed likelihood.

Why MLE is hard

For a mixture model,

$$Z_i \sim \text{ Categorical } (\pi_1, \ldots, \pi_K), \quad X_i \mid Z_i = k \sim f(X_i; \eta_k)$$

$$\ell(\theta) = \sum_i \log \left\{ \sum_k \pi_k f(x_i; \eta_k) \right\}.$$

Can we move the sum outside the logarithm?

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$$\log\left(\sum_k \pi_k f_k\right) \neq \sum_k \pi_k \log f_k.$$

The log of a sum couples the component parameters:

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The log of a sum couples the component parameters:

$$\frac{\partial \ell}{\partial \eta_k} = \sum_i r_{ik}(\theta) \frac{\partial \log f\left(x_i; \eta_k\right)}{\partial \eta_k}, \quad r_{ik}(\theta) = \frac{\pi_k f\left(x_i; \eta_k\right)}{\sum_j \pi_j f\left(x_i; \eta_j\right)}$$

If labels were known

Write $z_{ik} = \mathbf{1}\{Z_i = k\}$. Then

$$\ell_c(\theta) = \sum_k \sum_i z_{ik}\{\log \pi_k + \log f(x_i, \eta_k)\}.$$

- Fit each component η_k using its labeled observations.
- But the labels are unobserved.

If labels were known

Write $z_{ik} = \mathbf{1}\{Z_i = k\}$. Then

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- Fit each component η_k using its labeled observations.
- But the labels are unobserved.

Could we replace each label by a probability?

Conditional expectation

Define conditional expectation:

$$\mathbb{E}_{p_\theta(z|x)}[g(x,z)] = \int p_\theta(z \mid x) g(x,z)\, dz.$$

- Average over z.
- Hold x and θ fixed.

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For a latent component label,

$$\mathbb{E}_{p_\theta(z_i|x_i)}[\mathbf{1}\{z_i = k\}] = P_\theta(Z_i = k \mid x_i).$$

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Let $\theta^{(t)}$ be the current parameter estimate, define

$$\mathbb{E}_t[g(x,Z)] := \int p_{\theta^{(t)}}(z \mid x) g(x,z)\, dz.$$

Convex and concave

Convex

$$h(ta + (1-t)b) \leq th(a) + (1-t)h(b).$$

Example: $h(u) = u^2$.

Here $0 \leq t \leq 1$.

Concave

$$h(ta + (1-t)b) \geq th(a) + (1-t)h(b).$$

Example: $h(u) = \log u$.

The negative of a convex function is concave.

Jensen's inequality

$$h(\mathbb{E}[Y]) \leq \mathbb{E}[h(Y)] \qquad \text{if } h \text{ is convex,}$$

$$h(\mathbb{E}[Y]) \geq \mathbb{E}[h(Y)] \qquad \text{if } h \text{ is concave.}$$

Since log is strictly concave,

$$\boxed{\mathbb{E}[\log Y] \leq \log \mathbb{E}[Y]}, \qquad Y > 0.$$

Equality requires Y to be constant almost surely.

Split the joint density

For any candidate θ,

$$p_\theta(x, z) = p_\theta(x) p_\theta(z \mid x).$$

Take logarithms, then average using $\mathbb{E}_t$.

Split the joint density

For any candidate θ,

$$p_\theta(x, z) = p_\theta(x) p_\theta(z \mid x).$$

Take logarithms, then average using $\mathbb{E}_t$.

$$\log p_\theta(x, z) = \log p_\theta(x) + \log p_\theta(z \mid x).$$

Average both sides

$$\mathbb{E}_t[\log p_\theta(x, Z)] = \mathbb{E}_t[\log p_\theta(x)] + \mathbb{E}_t[\log p_\theta(Z \mid x)].$$

Which term does not depend on Z?

Average both sides

$$\mathbb{E}_t[\log p_\theta(x, Z)] = \mathbb{E}_t[\log p_\theta(x)] + \mathbb{E}_t[\log p_\theta(Z \mid x)].$$

Which term does not depend on Z?

$$\mathbb{E}_t[\log p_\theta(x)] = \log p_\theta(x) = \ell(\theta).$$

The expectation averages over Z, not over the observed data.

Define Q and H

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_t[\log p_\theta(x, Z)],$$

$$H(\theta \mid \theta^{(t)}) = \mathbb{E}_t[\log p_\theta(Z \mid x)].$$

Define Q and H

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_t[\log p_\theta(x, Z)],$$

$$H(\theta \mid \theta^{(t)}) = \mathbb{E}_t[\log p_\theta(Z \mid x)].$$

$$\boxed{Q(\theta \mid \theta^{(t)}) = \ell(\theta) + H(\theta \mid \theta^{(t)})}.$$

The expected complete log-likelihood splits into two terms.

Compare old and new

Use the same expectation at θ and $\theta^{(t)}$:

$$\ell(\theta) - \ell(\theta^{(t)}) = Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}) - \{H(\theta \mid \theta^{(t)}) - H(\theta^{(t)} \mid \theta^{(t)})\}.$$

That is:

$$\text{Likelihood change} = Q \text{ change} - H \text{ change}.$$

Or:

$$\Delta\ell = \Delta Q - \Delta H.$$

Prove $\Delta H \leq 0$

Assume common positive support and finite expectations.

$$H(\theta \mid \theta^{(t)}) - H(\theta^{(t)} \mid \theta^{(t)}) = \mathbb{E}_t\left[\log \frac{p_\theta(Z \mid x)}{p_{\theta^{(t)}}(Z \mid x)}\right].$$

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What is the expectation of the ratio inside the logarithm?

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What is the expectation of the ratio inside the logarithm?

$$\begin{aligned}
\mathbb{E}_t\left[\log \frac{p_\theta(Z \mid x)}{p_{\theta^{(t)}}(Z \mid x)}\right] &\leq \log \mathbb{E}_t\left[\frac{p_\theta(Z \mid x)}{p_{\theta^{(t)}}(Z \mid x)}\right] \\
&= \log \int p_{\theta^{(t)}}(z \mid x) \frac{p_\theta(z \mid x)}{p_{\theta^{(t)}}(z \mid x)}\, dz. \\
&= \log \int p_\theta(z \mid x)\, dz \\
&= 0
\end{aligned}$$

The key EM inequality

$$\Delta\ell = \Delta Q - \Delta H, \qquad \Delta H \le 0.$$

$$\boxed{\ell(\theta) - \ell(\theta^{(t)}) \ge Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)})}.$$

The key EM inequality

$$\Delta \ell = \Delta Q - \Delta H, \qquad \Delta H \leq 0.$$

$$\boxed{\ell(\theta) - \ell(\theta^{(t)}) \geq Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)})}.$$

Suppose $Q(\theta \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)})$, then

$$\ell(\theta) - \ell(\theta^{(t)}) \geq 0.$$

Increasing Q is enough to increase the observed likelihood.

E-step: average

Using the current conditional distribution, compute

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{p_{\theta^{(t)}}(z|x)}[\log p_\theta(x, z)].$$

- Average over uncertainty in the latent variables.
- Often only conditional moments are needed.

M-step: maximize

Hold the averaging distribution fixed and choose

$$\theta^{(t+1)} \in \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

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At the next iteration, recompute the conditional distribution.

Average under the old parameter; optimize over the new one.

Why every step ascends

$$\ell(\theta^{(t+1)}) - \ell(\theta^{(t)}) = \underbrace{Q(\theta^{(t+1)} \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)})}_{\text{M-step: nonnegative}} + \underbrace{H(\theta^{(t)} \mid \theta^{(t)}) - H(\theta^{(t+1)} \mid \theta^{(t)})}_{\text{Jensen: nonnegative}}.$$

Why every step ascends

$$\ell(\theta^{(t+1)}) - \ell(\theta^{(t)}) = \underbrace{Q(\theta^{(t+1)} \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)})}_{\text{M-step: nonnegative}} + \underbrace{H(\theta^{(t)} \mid \theta^{(t)}) - H(\theta^{(t+1)} \mid \theta^{(t)})}_{\text{Jensen: nonnegative}}.$$

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})}.$$

At a converged solution

Under regularity, the observed score satisfies

$$\begin{aligned}
\nabla_\theta \ell(\theta) &= \frac{\nabla_\theta p_\theta(x)}{p_\theta(x)} \\
&= \frac{\int \nabla_\theta p_\theta(x, z)\, dz}{p_\theta(x)} \\
&= \int \frac{p_\theta(x, z)}{p_\theta(x)} \frac{\nabla_\theta p_\theta(x, z)}{p_\theta(x, z)}\, dz \\
&= \int p_\theta(z \mid x)\, \nabla_\theta \log p_\theta(x, z)\, dz \\
&= \mathbb{E}_{p_\theta(z|x)}[\nabla_\theta \log p_\theta(x, Z)]\,.
\end{aligned}$$

At a converged solution

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&= \int p_\theta(z \mid x)\, \nabla_\theta \log p_\theta(x, z)\, dz \\
&= \mathbb{E}_{p_\theta(z|x)}[\nabla_\theta \log p_\theta(x, Z)]\,.
\end{aligned}$$

Equivalently,

$$\nabla_\theta \ell(\theta) = \nabla_\vartheta Q(\vartheta \mid \theta)|_{\vartheta=\theta}\,.$$

EM and stationarity

An interior exact M-step satisfies

$$\nabla_{\vartheta} Q(\vartheta \mid \theta^{(t)})\Big|_{\vartheta=\theta^{(t+1)}} = 0.$$

EM and stationarity

An interior exact M-step satisfies

$$\nabla_{\vartheta} Q(\vartheta \mid \theta^{(t)})\Big|_{\vartheta=\theta^{(t+1)}} = 0.$$

If $\theta^{(t)} \to \theta^{\star}$ and the gradient is continuous,

$$\nabla \ell(\theta^{\star}) = 0.$$

A regular converged solution is stationary, not necessarily a global MLE.

Gaussian mixture model

$$Z_i \sim \text{Categorical}(\pi_1, \ldots, \pi_K),$$
$$X_i \mid Z_i = k \sim \mathcal{N}(\mu_k, \sigma_k^2).$$

$$p_\theta(x_i) = \sum_k \pi_k \phi(x_i; \mu_k, \sigma_k^2).$$

Unknown weights, means, variances—and unobserved labels.

Complete log-likelihood

Let $z_{ik} = \mathbf{1}\{Z_i = k\}$. Up to a constant,

$$\ell_c(\theta) = \sum_{i,k} z_{ik} \left\{ \log \pi_k - \frac{1}{2} \log \sigma_k^2 - \frac{(x_i - \mu_k)^2}{2\sigma_k^2} \right\}.$$

Which quantities must the E-step average?

Complete log-likelihood

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Which quantities must the E-step average?

Only the latent indicators z_{ik}.

E-step: responsibilities

By Bayes' rule,

$$r_{ik}^{(t)} = \frac{\pi_k^{(t)} \phi(x_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)})}{\sum_j \pi_j^{(t)} \phi(x_i; \mu_j^{(t)}, (\sigma_j^2)^{(t)})}.$$

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$$r_{ik}^{(t)} = \mathbb{E}_t[z_{ik}], \qquad \sum_k r_{ik}^{(t)} = 1.$$

Responsibilities are soft assignments, not hard cluster labels.

Q is a weighted fit

Up to a constant,

$$Q(\theta \mid \theta^{(t)}) = \sum_{i,k} r_{ik}^{(t)} \left\{ \log \pi_k - \frac{1}{2} \log \sigma_k^2 - \frac{(x_i - \mu_k)^2}{2\sigma_k^2} \right\}.$$

Define effective component sizes:

$$N_k^{(t)} = \sum_i r_{ik}^{(t)}.$$

From now on, the responsibilities are fixed weights.

Update the weights

Maximize $\sum_k N_k^{(t)} \log \pi_k$ subject to $\sum_k \pi_k = 1$.

$$\mathcal{L} = \sum_k N_k^{(t)} \log \pi_k + \lambda \left(1 - \sum_k \pi_k \right).$$

Set the derivatives to zero and find λ.

Update the weights

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Set the derivatives to zero and find λ.

$$\frac{N_k^{(t)}}{\pi_k} = \lambda, \qquad \lambda = n, \qquad \boxed{\pi_k^{(t+1)} = \frac{N_k^{(t)}}{n}}.$$

Update the means

For component k,

$$\frac{\partial Q}{\partial \mu_k} = \frac{1}{\sigma_k^2} \sum_i r_{ik}^{(t)} (x_i - \mu_k).$$

Set this derivative to zero.

Update the means

For component k,

$$\frac{\partial Q}{\partial \mu_k} = \frac{1}{\sigma_k^2} \sum_i r_{ik}^{(t)} (x_i - \mu_k).$$

Set this derivative to zero.

$$\boxed{\mu_k^{(t+1)} = \frac{\sum_i r_{ik}^{(t)} x_i}{N_k^{(t)}}}.$$

A weighted average of the observations.

Update the variances

Let $v_k = \sigma_k^2$ and $S_k = \sum_i r_{ik}^{(t)} (x_i - \mu_k^{(t+1)})^2$.

$$Q_k(v_k) = -\frac{N_k^{(t)}}{2} \log v_k - \frac{S_k}{2v_k} + C.$$

Differentiate with respect to v_k.

Update the variances

Let $v_k = \sigma_k^2$ and $S_k = \sum_i r_{ik}^{(t)} (x_i - \mu_k^{(t+1)})^2$.

$$Q_k(v_k) = -\frac{N_k^{(t)}}{2} \log v_k - \frac{S_k}{2v_k} + C.$$

Differentiate with respect to v_k.

$$Q_k'(v_k) = \frac{S_k - N_k^{(t)} v_k}{2v_k^2} = 0 \quad \Longrightarrow \quad v_k = \frac{S_k}{N_k^{(t)}}.$$

The variance formula

$$\boxed{(\sigma_k^2)^{(t+1)} = \frac{\sum_i r_{ik}^{(t)} (x_i - \mu_k^{(t+1)})^2}{N_k^{(t)}}}.$$

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- Use the new mean.
- Keep the current responsibilities.
- Divide by $N_k^{(t)}$, not $N_k^{(t)} - 1$.

The fitting loop

1. Initialize valid weights, means, and variances.
2. Compute responsibilities.
3. Update weighted counts, means, and variances.
4. Record the observed likelihood and parameters.
5. Repeat until changes are small.

In the notebook: stable log-sum-exp and a variance floor of 0.01.

A Python demonstration

Simulation: $n = 500$

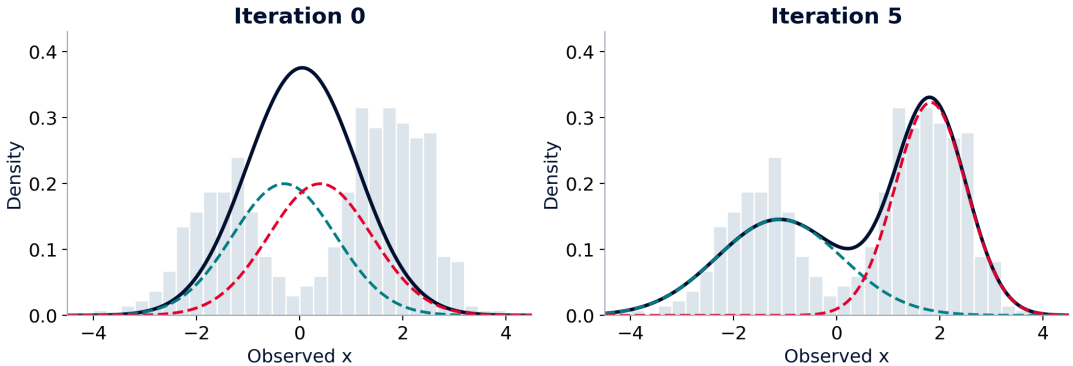
$$\pi = (0.4, 0.6),$$
$$\mu = (-1.5, 1.8),$$
$$\sigma^2 = (0.49, 0.64).$$

Initialization

$$\pi^{(0)} = (0.5, 0.5),$$
$$\mu^{(0)} = (-0.3, 0.4),$$
$$(\sigma^2)^{(0)} = (1, 1).$$

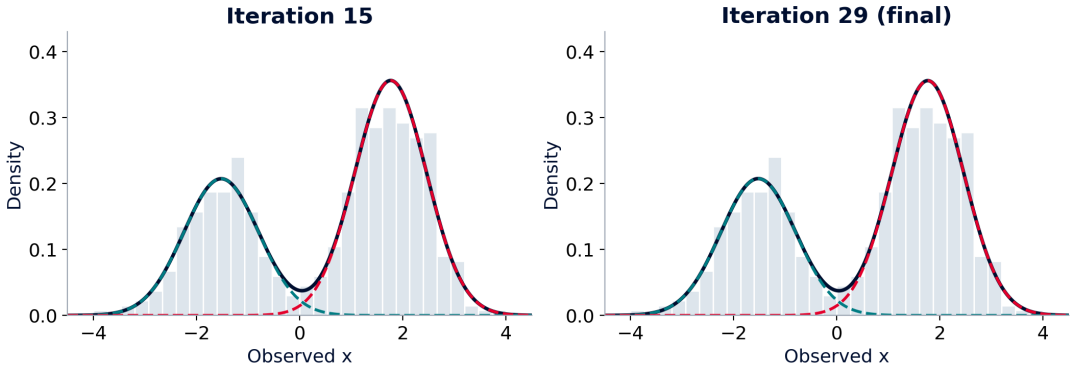
The fitting algorithm sees x_i, not the simulated labels.

The first few iterations



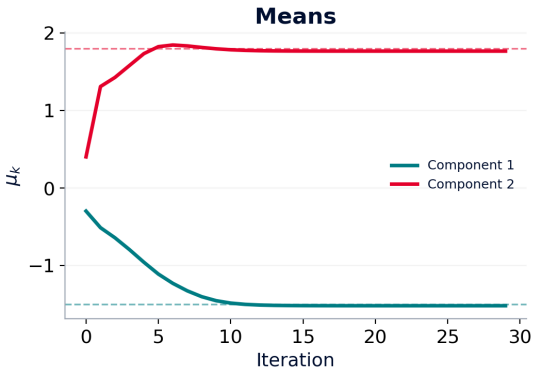
Overlapping initial components begin to separate.

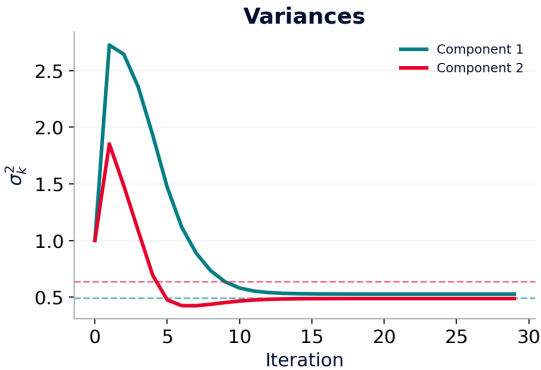
The fitted mixture



This run converges after 29 iterations.

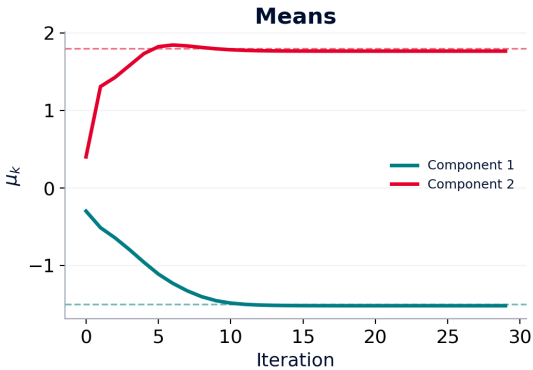
Means and variances

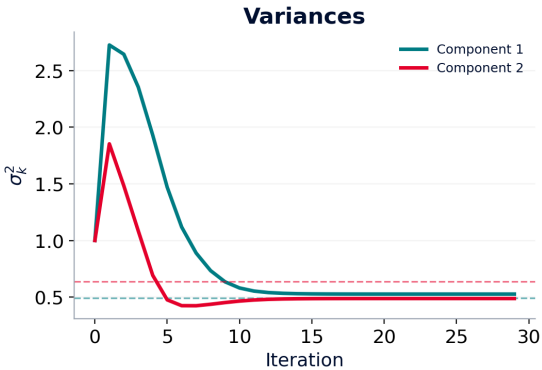




Do individual parameters have to move monotonically?

Means and variances

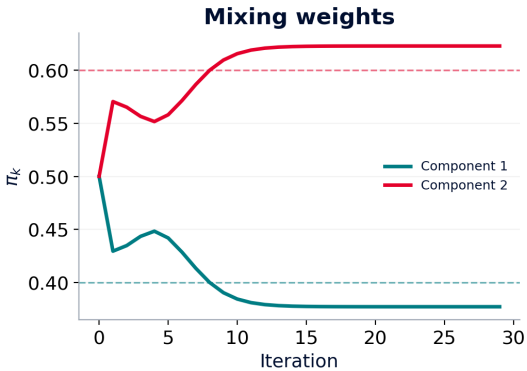


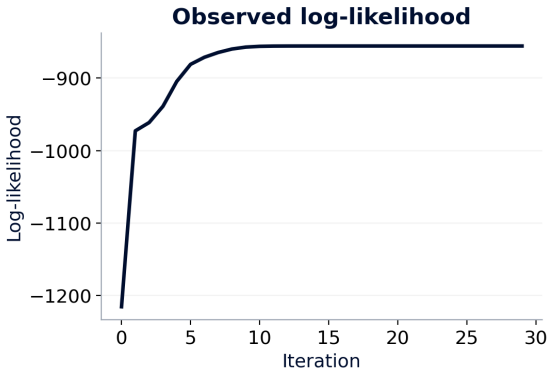


Do individual parameters have to move monotonically?

No. The ascent guarantee applies to the likelihood.

Weights and likelihood





$$\ell(\theta^{(0)}) \approx -1215.55 \quad \longrightarrow \quad \ell(\theta^{(29)}) \approx -855.62.$$

Generalized EM

When full maximization is expensive, just increase the exact Q:

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- For example: a few improving optimization steps.
- The usual likelihood ascent proof still applies.

GEM simplifies the M-step.

Monte Carlo EM

When the expectation is difficult, draw $z^{(m)} \sim p_{\theta^{(t)}}(z \mid x)$:

$$\widehat{Q}_t(\theta) = \frac{1}{M_t} \sum_{m=1}^{M_t} \log p_\theta(x, z^{(m)}).$$

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When the expectation is difficult, draw $z^{(m)} \sim p_{\theta^{(t)}}(z \mid x)$:

$$\widehat{Q}_t(\theta) = \frac{1}{M_t} \sum_{m=1}^{M_t} \log p_\theta(x, z^{(m)}).$$

- Maximize the simulated objective with the draws fixed.
- Monte Carlo error can break likelihood monotonicity.

MCEM approximates the E-step.

Three takeaways

1. Latent variables make the complete-data objective easier.
2. Conditional expectation plus Jensen gives

$$\Delta \ell \geq \Delta Q.$$

3. Gaussian mixture EM alternates soft assignments and weighted fits.

E-step: average. **M-step: improve.**